

SYSTEM REQUIREMENTS SPECIFICATION

CSI-100 Cell Sorting Instrument (IVD)

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Purpose

This specification defines the system-level functional, performance, safety and interface requirements for the CSI-100. It captures stakeholder and user needs and the intended use, and serves as the design input baseline under 21 CFR 820 and ISO 13485:2016 clause 7.3.3.

Document Control

Rev	Date	Description of change	Author
A	Nov 2024	Initial release at Design Input Review 1 (concept freeze).	A. Lodhi
B	May 2025	Added 561 nm and 445 nm laser options and fourth beam spot; added configuration requirements; updated sort performance after feasibility data.	A. Lodhi
C	Jan 2026	Added aerosol management, plate deposition and temperature control options; added cybersecurity and data integrity requirements; aligned safety requirements with HA-CSI-001 Rev B.	A. Lodhi
D	Aug 2026	Design Input Review 3 (verification baseline). Tightened carryover and clog response requirements; added transport and installation requirements; completed traceability to user needs and verification documents for design transfer.	A. Lodhi

Approvals

Function	Role	Signature / Date
Systems Engineering	Author, System Requirements Owner	
Marketing	Product Requirements Owner	
R&D Engineering	Design Owner	
Software Engineering	Software Requirements Owner	
Quality Engineering	Design Quality	
Regulatory Affairs	Regulatory Strategy	
Clinical / Medical Affairs	Clinical Requirements	
Manufacturing Engineering	Manufacturability Review	
Service & Support	Serviceability Review	

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1. Introduction

1.1 Purpose

This System Requirements Specification (SRS) defines the system-level functional, performance, safety, interface and compliance requirements for the CSI-100 Cell Sorting Instrument. It translates the intended use, stakeholder needs and user needs into verifiable design inputs, as required by 21 CFR 820 (QMSR) and ISO 13485:2016 clause 7.3.3.

This document is the foundation of the CSI-100 design control process. Every subsystem design specification, software requirement, design output, verification test and validation activity traces back to a requirement in this document, and every requirement in this document traces back to a user need, a risk control, or an applicable standard.

1.2 Scope

This SRS applies to the complete CSI-100 system as delivered to the customer: the benchtop sorter, the fluidics supply module, the workstation and instrument control software, the accessory kit, and the optional aerosol evacuation module, automated plate deposition unit and collection temperature control. It applies to all released laser and detector configurations defined in CFG-CSI-001.

IVD reagents, assays and setup or QC beads are outside the scope of this document; their interface to the instrument is defined here only where the instrument must support them. Detailed software requirements are in SRS-SW-CSI-010, and detailed subsystem requirements are in the subsystem design specifications (SSD-CSI-100 to SSD-CSI-950), each of which traces to this document.

1.3 Requirement conventions

Each requirement has a unique identifier of the form SYS-nnn, where the first digit matches the section of this document. Requirement identifiers are never reused; deleted requirements are marked as deleted and retained. The word **shall** indicates a mandatory requirement that will be verified. The word **should** indicates a goal that is not formally verified. Notes and rationale are informative.

Each requirement table gives the following attributes:

Attribute	Meaning
Source	Origin of the requirement: a user need (UN-nn, Section 2.6), a hazard or risk control (HA-nn, from HA-CSI-001), a product requirement (PRD, from PRD-CSI-001) or an external standard
Allocation	Subsystem that implements the requirement (SS-01 to SS-10, Section 2.4). SYS means the requirement is met by the integrated system
Verification	Planned verification method: T = test, I = inspection, A = analysis, D = demonstration. The verification document for each section is listed in Section 14

1.4 Requirement change control

This document is controlled under the document control procedure. Changes after Design Input Review 3 require an Engineering Change Order with an impact assessment covering downstream specifications, the risk management file, verification status and the design transfer trace matrix (TM-CSI-DT-01).

2. Product Overview and User Needs

2.1 Intended use

The CSI-100 Cell Sorting Instrument is an automated, multi-laser flow cytometer and droplet cell sorter intended for the identification, enumeration and physical separation of cell populations in human specimens (such as peripheral blood, bone marrow and cellular therapy products) stained with fluorochrome-labeled reagents. It is intended for in vitro diagnostic use by trained laboratory professionals in clinical laboratories, when used with IVD reagents and assays validated for the instrument. Sorted cells are recovered for further laboratory analysis.

The instrument does not itself provide a diagnosis. Results are interpreted by qualified laboratory personnel together with other clinical and laboratory findings. The instrument is not intended for point-of-care or home use, or for returning sorted cells to a patient.

2.2 Users and use environment

The intended operators are clinical laboratory scientists trained on the instrument by the manufacturer or a trained trainer. Installation, alignment verification, repair and preventive maintenance are performed by manufacturer-trained service engineers. Laboratory supervisors and administrators manage user accounts, assays and quality control review.

The instrument is used on a laboratory bench in a clinical laboratory with standard electrical service and no special room requirements. Specimens may be potentially infectious, so the user facility performs a biosafety assessment for sorting and may operate the instrument inside a biosafety enclosure or with the aerosol evacuation option.

2.3 Principle of operation

Pressurized sheath fluid from a mobile fluidics supply module hydrodynamically focuses the sample into a single-file core stream inside a quartz cuvette flow cell. Up to four fiber-delivered laser beams, focused as spatially separated spots, cross the stream. Scattered light and fluorescence are collected by a gel-coupled objective and routed by collection fibers to detector arrays, where dichroic mirrors and bandpass filters direct each wavelength band to a photomultiplier tube. Digitized pulses are compared with user-defined gates in real time. Below the nozzle the stream breaks into droplets; a droplet containing a target cell is charged at breakoff and deflected by high-voltage plates into a collection tube, plate well or slide position, while other droplets go to waste.

2.4 System context and architecture

The system consists of the benchtop sorter, the fluidics supply module connected by an umbilical, and a workstation connected by Ethernet. Requirements are allocated to the ten subsystems shown in Figure 1, which are the same subsystems used in the risk management file and the design transfer plan.

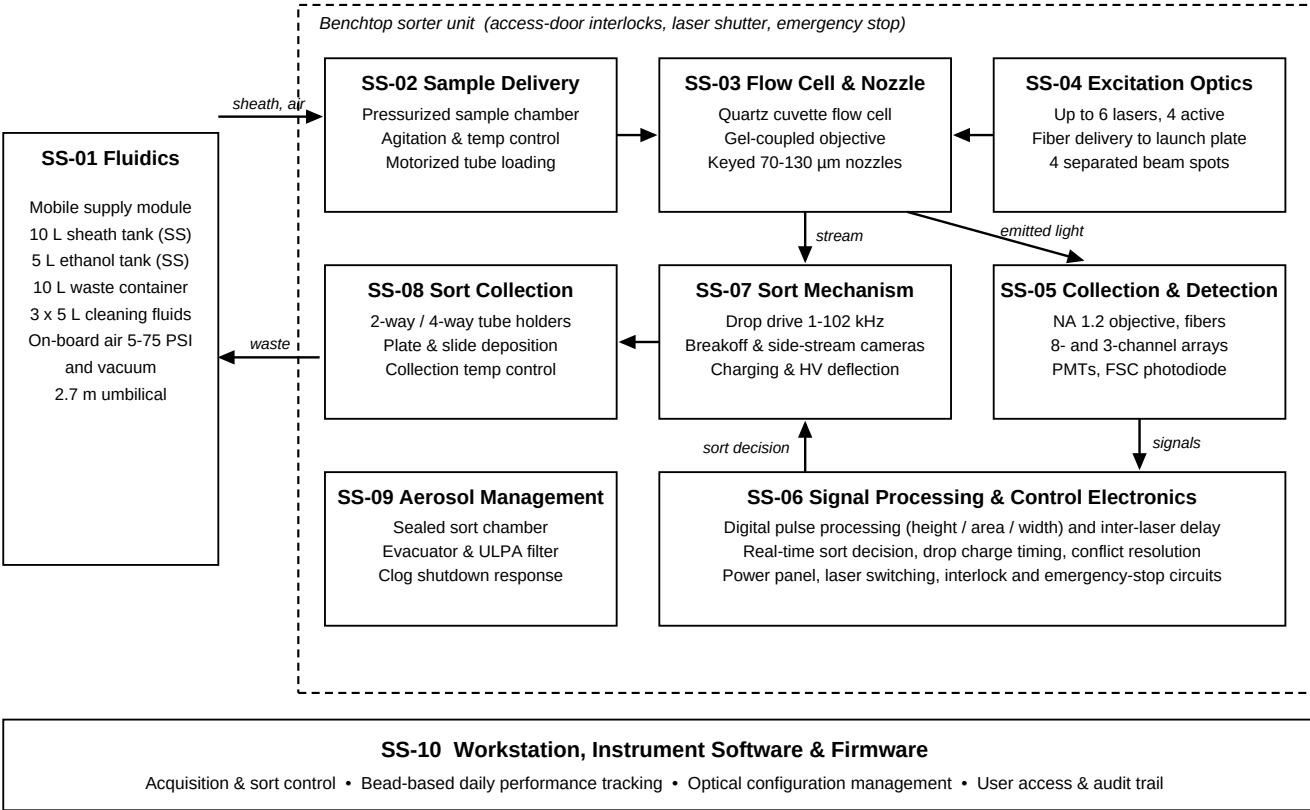


Figure 1. CSI-100 system architecture and subsystem allocation (SS-01 to SS-10).

ID	Subsystem	Governing design specification
SS-01	Fluidics supply module	SSD-CSI-100
SS-02	Sample delivery	SSD-CSI-100
SS-03	Flow cell and nozzle	SSD-CSI-300
SS-04	Excitation optics	SSD-CSI-400
SS-05	Collection and detection	SSD-CSI-500
SS-06	Signal processing and control electronics	SSD-CSI-600
SS-07	Sort mechanism	SSD-CSI-700
SS-08	Sort collection and plate deposition	SSD-CSI-800
SS-09	Aerosol management	SSD-CSI-900
SS-10	Workstation and instrument control software	SRS-SW-CSI-010

2.5 Stakeholders

Stakeholder	Primary needs
Patients (indirect)	Accurate results; sorted cells correctly identified and not cross-contaminated
Clinical laboratory scientists	Reliable, repeatable performance; guided setup; safe operation; minimal manual adjustment
Laboratory directors and supervisors	Throughput, QC traceability, data integrity, user access control, regulatory compliance
Biosafety officers	Containment of aerosols from infectious specimens; safe clog handling; decontamination
Service engineers	Access for maintenance, diagnostics, field-replaceable modules, stable alignment
Manufacturing	Producible design, testable characteristics, configuration control
Regulatory Affairs	Compliance with US and EU IVD regulations and harmonized standards

2.6 User needs

User needs are derived from PRD-CSI-001, customer interviews, competitive analysis and the use-related risk analysis. They are validated in design validation (DVAL-CSI-001); the system requirements that satisfy each need are verified in design verification.

ID	User need
UN-01	Identify and enumerate cell populations in stained human specimens with results accurate and precise enough for clinical use.
UN-02	Physically separate target cell populations at high purity and recovery, with cells remaining viable for downstream analysis.
UN-03	Measure many markers on each cell at the same time (multicolor panels).
UN-04	Process specimens at a throughput that fits the laboratory workday.
UN-05	Collect sorted cells into tubes, multiwell plates and slides, including single-cell deposition with a record of each deposited cell.
UN-06	Operate the instrument without risk of injury from laser light, high voltage, moving parts, pressure or biohazardous aerosols.
UN-07	Install the instrument in a standard clinical laboratory without special facilities.
UN-08	Obtain consistent performance from day to day with minimal user adjustment, confirmed by a daily quality control check.
UN-09	Set up, run, clean and shut down the instrument through guided, largely automated procedures.
UN-10	Keep results and records secure, attributable and traceable, with controlled user access.
UN-11	Keep samples and sorted cells at a chosen temperature during the run.
UN-12	Keep the instrument in service with predictable maintenance and fast repair.
UN-13	Use the instrument in the United States and the European Union in compliance with IVD regulations.
UN-14	Avoid carryover and cross-contamination between specimens.
UN-15	Handle cells of different sizes and fragility.
UN-16	Choose the laser and detector configuration that fits the laboratory's assays, and upgrade it later.

3. Physical, Electrical and Environmental Requirements

3.1 Physical characteristics

ID	Requirement	Source	Allocation	Verification
SYS-301	The sorter shall not exceed 71 cm height, 122 cm width and 71 cm depth (28 × 48 × 28 in), excluding the workstation.	UN-07	SYS	I
SYS-302	The sorter shall weigh no more than 185 kg (408 lb) in its heaviest configuration.	UN-07	SYS	I
SYS-303	The fluidics supply module shall not exceed 66 cm height, 81 cm width and 66 cm depth (26 × 32 × 26 in), and shall weigh no more than 82 kg (180 lb) without tanks.	UN-07	SS-01	I
SYS-304	The fluidics supply module shall be mobile on lockable casters with leveling feet.	UN-07	SS-01	I
SYS-305	The sorter shall operate on a laboratory bench or on an optional bench no larger than 85.1 × 266.7 × 134.6 cm.	UN-07	SYS	I
SYS-306	The umbilical between sorter and fluidics supply module shall allow separation of up to 2.7 m (9 ft).	UN-07	SS-01	I

3.2 Electrical requirements

ID	Requirement	Source	Allocation	Verification
SYS-311	The system shall operate from 100, 115 or 230 VAC (±10%), 50 or 60 Hz, single phase.	UN-07	SS-06	T
SYS-312	Total power consumption of the sorter and fluidics supply module shall not exceed 1,500 W.	UN-07	SYS	T
SYS-313	The system shall require no more than one 20 A outlet for the sorter and fluidics supply module combined.	UN-07	SYS	A
SYS-314	The fluidics supply module shall be powered and switched only from the sorter.	HA-12	SS-01, SS-06	I
SYS-315	The system shall be protected by a resettable circuit breaker.	HA-12	SS-06	I
SYS-316	The system shall meet installation category II and pollution degree 2 as defined in IEC 61010-1.	IEC 61010-1	SYS	A

3.3 Environmental requirements

ID	Requirement	Source	Allocation	Verification
SYS-321	The system shall meet all performance requirements at ambient temperatures of 15 to 30 °C.	UN-07	SYS	T
SYS-322	The system shall meet all performance requirements at 5 to 80% relative humidity, non-condensing.	UN-07	SYS	T
SYS-323	The system shall operate at altitudes up to 2,000 m.	IEC 61010-1	SYS	A
SYS-324	The system shall withstand storage at 1 to 40 °C without loss of performance.	UN-07	SYS	T
SYS-325	The system shall require no special room facilities, no external compressed air and no external vacuum.	UN-07	SS-01	D
SYS-326	The system shall accept optional facility air at 80 to 100 PSI, dry and oil-free.	UN-07	SS-01	T

3.4 Acoustic noise

ID	Requirement	Source	Allocation	Verification
SYS-331	Acoustic noise at the operator position shall not exceed 62 dBA during sorting.	UN-07	SYS	T

4. Performance Requirements

Unless stated otherwise, performance requirements apply to every released configuration, with the 70 µm nozzle at its default sort setup, after the laser warm-up period and a passing daily QC check.

4.1 Fluorescence resolution

ID	Requirement	Source	Allocation	Verification
SYS-401	The coefficient of variation (CV) of the full G0/G1 peak for propidium iodide-stained chicken erythrocyte nuclei, excited at 488 nm, shall be less than 3.0% (area parameter).	UN-01	SS-03, SS-04, SS-05	T
SYS-402	The CV of the full G0/G1 peak for Hoechst-stained chicken erythrocyte nuclei, excited at 405 nm or 375 nm, shall be less than 3.5% (area parameter).	UN-01	SS-03, SS-04, SS-05	T

4.2 Fluorescence linearity

ID	Requirement	Source	Allocation	Verification
SYS-411	The doublet-to-singlet ratio for chicken erythrocyte nuclei shall be 1.95 to 2.05, for PI excited at 488 nm and for Hoechst excited at 405 nm or 375 nm.	UN-01	SS-05, SS-06	T

4.3 Scatter sensitivity

ID	Requirement	Source	Allocation	Verification
SYS-421	Forward and side scatter shall resolve 0.5 µm beads from instrument noise.	UN-01	SS-04, SS-05	T
SYS-422	Scatter sensitivity shall allow separation of fixed platelets from noise and identification of bacteria.	UN-01	SS-04, SS-05	T
SYS-423	The forward scatter path shall accept interchangeable neutral density filters of 0.5, 1.0, 1.5 and 2.0 for small and large particles.	UN-15	SS-05	I

4.4 Excitation

ID	Requirement	Source	Allocation	Verification
SYS-431	The system shall support up to six installed lasers, with up to four active at the same time.	UN-03, UN-16	SS-04	D
SYS-432	The 488 nm and 633 nm lasers shall be standard in every configuration. 375, 405, 445 and 561 nm lasers shall be available as options.	UN-03, UN-16	SS-04	I
SYS-433	Minimum laser power out of the laser head shall be: 375 nm > 7 mW; 405 nm > 50 mW; 445 nm > 20 mW; 488 nm > 20 mW; 561 nm > 50 mW; 633 nm > 18 mW.	UN-01	SS-04	T
SYS-434	Elliptical beams at the interrogation point shall measure $9 \pm 3 \mu\text{m}$ in height and $65 \pm 7 \mu\text{m}$ in width.	UN-01	SS-04	T
SYS-435	Each active laser shall be focused on a spatially separated beam spot. Lasers sharing a spot (488/445 and 405/375) shall be mutually exclusive.	UN-03	SS-04, SS-06	T
SYS-436	Optical alignment shall be fixed at the factory and shall require no user adjustment.	UN-08	SS-04	D
SYS-437	The system shall reach full performance within 30 minutes of laser power-on.	UN-08, UN-09	SS-04	T
SYS-438	Lasers shall be air-cooled and shall need no external cooling or special power.	UN-07	SS-04	I

4.5 Detection

ID	Requirement	Source	Allocation	Verification
SYS-441	The system shall measure forward scatter, side scatter and at least 18 fluorescence parameters simultaneously in the fully populated configuration.	UN-03	SS-05	T
SYS-442	Each fluorescence detector shall have a user-replaceable bandpass filter and each detector array shall use user-replaceable longpass dichroic mirrors.	UN-16	SS-05	I
SYS-443	The system shall report height, area and width for every detected parameter.	UN-01	SS-06	T
SYS-444	Signals from each beam spot shall be time-aligned to the same event, with inter-laser delay set automatically during daily QC.	UN-01, UN-08	SS-06, SS-10	T
SYS-445	Data shall be digitized with a linear dynamic range of at least 18 bits.	UN-01	SS-06	A

4.6 Acquisition rate

ID	Requirement	Source	Allocation	Verification
SYS-451	The system shall acquire at least 70,000 events per second with 8 parameters and 12 compensation pairs without loss of data.	UN-04	SS-06	T
SYS-452	The system shall acquire more than 100,000 events per second when fewer parameters are recorded.	UN-04	SS-06	T
SYS-453	Sample flow rate shall be user-selectable over at least 10 to 80 $\mu\text{L}/\text{min}$.	UN-04	SS-02	T

4.7 Sort performance

ID	Requirement	Source	Allocation	Verification
SYS-461	For a four-way sort at 70 PSI, 90 kHz drop drive and an average threshold rate of 25,000 events per second, sort purity shall exceed 98% and yield shall exceed 80% of Poisson expected yield.	UN-02	SS-06, SS-07	T
SYS-462	The system shall sort into up to four populations at the same time.	UN-02	SS-07	D
SYS-463	Drop drive frequency shall be adjustable from 1 to 100 kHz or wider.	UN-15	SS-07	T
SYS-464	The system shall resolve particle position within the drop to 1/32 of a drop and shall provide user-selectable purity, yield and single-cell sort modes.	UN-02, UN-05	SS-06, SS-10	T
SYS-465	The system shall automatically determine drop delay using fluorescent drop-delay beads.	UN-08, UN-09	SS-07, SS-10	T
SYS-466	Sorted cell lines and human PBMCs shall remain viable and proliferate after sorting at every default sheath pressure.	UN-02	SYS	T
SYS-467	The system shall provide nozzles of 70, 85 and 100 μm , with a 130 μm nozzle as an option, each with a default sort setup.	UN-15	SS-03, SS-07	I
SYS-468	Changing the nozzle shall not require optical realignment.	UN-08	SS-03	D

4.8 Carryover

ID	Requirement	Source	Allocation	Verification
SYS-471	Sample carryover from one tube to the next shall not exceed 0.5% after the automatic sample line flush.	UN-14	SS-02	T
SYS-472	The system shall flush the inside and outside of the sample line with sheath after every tube and shall provide a sample line backflush on demand.	UN-14	SS-02, SS-10	D

4.9 Sort collection

ID	Requirement	Source	Allocation	Verification
SYS-481	The system shall provide two-way sorting into 12 $\times$ 75 mm and 15 mL tubes, and four-way sorting into 12 $\times$ 75 mm tubes, 1.5 mL tubes and 1 mL microtubes.	UN-05	SS-08	D
SYS-482	With the plate deposition option, the system shall sort into 6-, 24-, 48-, 96- and 384-well plates and onto microscope slides.	UN-05	SS-08	D
SYS-483	With the plate deposition option, the system shall deposit single cells so that at least 95% of target wells receive exactly one particle.	UN-05	SS-07, SS-08	T
SYS-484	The system shall record the measured parameters of each cell deposited in a plate well (index sorting).	UN-05	SS-10	D
SYS-485	The system shall allow users to define custom collection devices.	UN-05	SS-10	D

4.10 Fluidics

ID	Requirement	Source	Allocation	Verification
SYS-491	Sheath pressure shall be software-controlled from 5 to 75 PSI.	UN-15	SS-01	T
SYS-492	The fluidics supply module shall hold at least 10 L of sheath, 10 L of waste, 5 L of shutdown ethanol and three 5 L cleaning fluids.	UN-04, UN-09	SS-01	I
SYS-493	The system shall warn the user before sheath runs out or waste is full, and shall prevent sorting when either condition would interrupt a sort.	UN-09, HA-15	SS-01, SS-10	T
SYS-494	The sample chamber shall accept 15 mL, 12 × 75 mm and 1 mL tubes.	UN-04	SS-02	I
SYS-495	The system shall agitate the sample tube at user-selectable speeds.	UN-01	SS-02	D

4.11 Temperature control

ID	Requirement	Source	Allocation	Verification
SYS-4101	The sample chamber shall hold a user-selected temperature, reached within 45 minutes with the chamber closed.	UN-11	SS-02	T
SYS-4102	With the temperature control option, collection tubes shall be held at a user-selected temperature within ± 2 °C.	UN-11	SS-08	T

5. Safety Requirements

Safety requirements implement risk controls from the hazard analysis (HA-CSI-001). Each HA reference identifies the hazard the requirement controls.

5.1 Electrical safety

ID	Requirement	Source	Allocation	Verification
SYS-501	The system shall comply with IEC 61010-1 and IEC 61010-2-101 for protection against electric shock and fire.	HA-12, IEC 61010	SYS	T
SYS-502	Protective earth continuity, dielectric strength and touch current shall be tested on every unit in production.	HA-12	SYS	T
SYS-503	The system shall comply with IEC 61326-1 and IEC 61326-2-6 for electromagnetic compatibility.	HA-13, IEC 61326	SYS	T

5.2 Laser safety

ID	Requirement	Source	Allocation	Verification
SYS-511	The system shall be a Class 1 laser product per IEC 60825-1 in all user-accessible conditions.	HA-01, IEC 60825-1	SYS	T
SYS-512	Opening the flow cell access door shall block all laser beams by a shutter before any beam is accessible.	HA-01	SS-04	T
SYS-513	Laser warning labels shall be placed on all service-access panels behind which Class 3B radiation is present.	HA-01	SYS	I

5.3 Interlocks and high voltage

ID	Requirement	Source	Allocation	Verification
SYS-521	The sort collection chamber door shall not open unless the sort block door is closed.	HA-05	SS-07	T
SYS-522	An illuminated warning light and a software indicator shall show whenever deflection plate voltage is on.	HA-02	SS-07, SS-10	T
SYS-523	Deflection plate voltage shall not exceed 12,000 V between plates and shall be insulated so that no energized part is accessible with the sort block door closed.	HA-02	SS-07	T
SYS-524	The system shall fully depressurize within 10 seconds of main power switch-off.	HA-09	SS-01	T

5.4 Emergency stop

ID	Requirement	Source	Allocation	Verification
SYS-531	The system shall provide an illuminated emergency stop button next to the sample loading port.	HA-06	SS-06	I
SYS-532	Pressing the emergency stop shall, within 2 seconds: unload the sample tube, stop plate stage motion, turn off the stream, turn off deflection plate voltage, close the waste aspirator drawer and display a warning.	HA-02, HA-06	SS-06, SS-10	T
SYS-533	The emergency stop button shall light while the sample loading port is moving.	HA-06	SS-06	T

5.5 Clog and breakoff fault response

ID	Requirement	Source	Allocation	Verification
SYS-541	The system shall monitor the drop breakoff continuously during sorting.	HA-04, HA-07	SS-07, SS-10	T
SYS-542	If the breakoff moves outside its tolerance, the system shall pause sorting until the breakoff recovers, while sample continues to flow.	HA-07	SS-07, SS-10	T
SYS-543	If a nozzle clog is detected, the system shall, within 2 seconds: stop the stream, turn off plate voltage, close the aspirator drawer and unload the sample tube.	HA-04	SS-07, SS-10	T

5.6 Biohazard containment

ID	Requirement	Source	Allocation	Verification
SYS-551	With the sort block and collection chamber doors closed and the aerosol evacuation option running, no aerosolized particles shall be detected outside the sort chamber during sorting or during a simulated clog.	HA-04, UN-06	SS-09	T
SYS-552	The aerosol evacuation option shall provide at least 7 CFM in normal mode and at least 30 CFM in boost mode through a ULPA filter with efficiency above 99.9995% at 0.12 µm.	HA-04	SS-09	T
SYS-553	All wetted surfaces and surfaces exposed to specimens shall withstand the decontamination agents specified in the IFU (10% bleach, 70% ethanol).	HA-10	SYS	T
SYS-554	The IFU shall describe safe handling of a clog and decontamination after sorting infectious specimens.	HA-04	SYS	I

5.7 Mechanical and pressure safety

ID	Requirement	Source	Allocation	Verification
SYS-561	Pressurized tanks shall be rated for at least 1.5 times the maximum operating pressure and shall include overpressure relief.	HA-09	SS-01	A, T
SYS-562	Moving parts (loading port, plate stage, aspirator drawer) shall not exert a pinching force greater than the limits of IEC 61010-1.	HA-06	SS-02, SS-08	T
SYS-563	The system shall remain stable when tilted 10° in any direction.	HA-11, IEC 61010-1	SYS	T

6. Functional Requirements

6.1 Startup, shutdown and cleaning

ID	Requirement	Source	Allocation	Verification
SYS-601	The system shall perform an automated fluidics startup that primes all fluid lines and removes air.	UN-09	SS-01, SS-10	D
SYS-602	The system shall perform an automated fluidics shutdown that leaves the flow path filled with 70% ethanol.	UN-09	SS-01, SS-10	D
SYS-603	The system shall provide automated cleaning modes: flow cell clean, sample line backflush, prime after tank refill, and a bleach, water and ethanol clean cycle.	UN-09, UN-14	SS-01, SS-10	D
SYS-604	The system shall be ready to run daily QC within 45 minutes of power-on, including laser warm-up.	UN-04	SYS	T

6.2 Sort setup and stream control

ID	Requirement	Source	Allocation	Verification
SYS-611	Selecting a nozzle size in software shall load a verified default setup for sheath pressure, drop drive, breakoff targets, drop delay, plate voltage, stream charges and laser delays.	UN-09, UN-15	SS-07, SS-10	D
SYS-612	The system shall display a live image of the drop breakoff and of the side streams.	UN-09	SS-07	D
SYS-613	The system shall hold the breakoff position automatically by adjusting drop drive amplitude.	UN-08	SS-07	T
SYS-614	The system shall provide a test sort function that generates side streams for adjustment without sample.	UN-09	SS-07	D

6.3 Quality control and configuration

ID	Requirement	Source	Allocation	Verification
SYS-621	The system shall run a daily bead-based performance check that measures detector performance, sets inter-laser delay and records results with pass/fail status.	UN-08	SS-10	T
SYS-622	The system shall not allow an IVD assay to run unless the most recent performance check for the active configuration has passed within the last 24 hours.	UN-08, HA-08	SS-10	T
SYS-623	The software configuration shall identify the installed lasers, mirrors and filters, and shall warn the user if the selected configuration does not match the active lasers.	UN-16, HA-08	SS-10	T
SYS-624	Performance check results shall be stored and trended so users can see changes over time.	UN-08	SS-10	D

6.4 Acquisition, analysis and sorting

ID	Requirement	Source	Allocation	Verification
SYS-631	The system shall allow users to define gates, sort populations and sort layouts for each collection device.	UN-02, UN-05	SS-10	D
SYS-632	The system shall calculate fluorescence compensation from single-stained controls.	UN-01, UN-03	SS-10	T
SYS-633	The system shall show live sort counters for each sort population and the sort efficiency.	UN-02	SS-10	D
SYS-634	The user shall be able to pause, resume and stop a sort without losing the sort record.	UN-02	SS-10	D
SYS-635	The system shall produce a sort report giving the sort settings, counts and events for each sort.	UN-10	SS-10	D

7. Software, Data Integrity and Cybersecurity Requirements

Detailed software requirements are in SRS-SW-CSI-010. The instrument control software is developed under IEC 62304 with software safety class B.

ID	Requirement	Source	Allocation	Verification
SYS-701	The software shall require a unique user login and support at least operator, supervisor and administrator roles with different permissions.	UN-10	SS-10	T
SYS-702	The software shall keep a secure, computer-generated, time-stamped audit trail of record creation, modification and deletion that users cannot edit, as required by 21 CFR Part 11.	UN-10, 21 CFR 11	SS-10	T
SYS-703	The software shall support electronic signatures for result release and assay approval.	UN-10, 21 CFR 11	SS-10	T
SYS-704	Approved IVD assay templates shall be locked so that operators cannot change instrument settings, gates or analysis once approved.	UN-01, HA-08	SS-10	T
SYS-705	The software shall export data in FCS 3.1 format and reports in PDF.	UN-10	SS-10	T
SYS-706	The software shall detect and report loss of communication with the instrument without corrupting stored data.	UN-10	SS-10	T
SYS-707	The workstation shall have a locked, validated operating system image, with only the services needed for operation enabled.	Cybersecurity	SS-10	I
SYS-708	The manufacturer shall maintain a software bill of materials and a process for assessing and patching vulnerabilities.	Cybersecurity	SS-10	I
SYS-709	User data at rest shall be protected by access control, and data exported off the workstation shall be integrity-protected.	UN-10, Cybersecurity	SS-10	T

8. Interface Requirements

Detailed interface definitions are in ICD-CSI-010 (sorter to fluidics supply module) and ICD-CSI-020 (sorter to workstation).

ID	Requirement	Source	Allocation	Verification
SYS-801	The sorter and fluidics supply module shall connect through a keyed umbilical carrying power, sheath, waste, air and a serial communication line; connectors shall not be interchangeable.	HA-14	SS-01	I
SYS-802	The sorter shall connect to the workstation by Ethernet on a dedicated, isolated network link.	UN-10	SS-06, SS-10	I
SYS-803	The system shall provide connections for the aerosol evacuation module and the temperature control option without tools.	UN-11, UN-06	SS-01, SS-09	D
SYS-804	Collection tube holders shall share a universal top with evacuation ports compatible with the aerosol evacuation option.	UN-06	SS-08	I
SYS-805	The system shall use sheath fluid of buffered saline or 1× PBS, and cleaning fluids of 10% bleach, deionized water and 70% ethanol.	UN-09	SS-01	T
SYS-806	The system shall support IVD reagents and QC, alignment and drop-delay beads as specified in ACC-CSI-001.	UN-01, UN-08	SYS	T

9. Usability Requirements

Usability engineering is performed under IEC 62366-1. Use-related hazards are analyzed in HA-CSI-001, and summative evaluation is reported in the usability engineering file.

ID	Requirement	Source	Allocation	Verification
SYS-901	A trained operator shall be able to complete startup, daily QC, a four-way sort and shutdown using only the IFU and on-screen guidance.	UN-09	SYS	D
SYS-902	All critical tasks identified in the use-related risk analysis shall be completed without use errors that could cause harm in summative usability testing.	UN-06, IEC 62366-1	SYS	T
SYS-903	Tube holders, nozzles and collection devices shall be marked with their size and keyed or shaped to prevent incorrect installation.	HA-08	SS-02, SS-03, SS-08	I
SYS-904	The software shall warn the user before any action that would discard unsaved data or stop an active sort.	UN-10	SS-10	D
SYS-905	Fluid containers shall be labeled and color-coded by content, and their connectors keyed to prevent misconnection.	HA-14	SS-01	I

10. Reliability, Maintainability and Serviceability Requirements

ID	Requirement	Source	Allocation	Verification
SYS-1001	The system shall have a design service life of at least 7 years with scheduled maintenance.	UN-12	SYS	A
SYS-1002	Mean time between failures requiring a service visit shall be at least 2,000 operating hours.	UN-12	SYS	A, T
SYS-1003	Routine user maintenance (filters, sample lines, waste caps, nozzle cleaning) shall be possible without tools other than those in the accessory kit, and shall not need realignment.	UN-12	SYS	D
SYS-1004	User maintenance intervals shall be no more frequent than weekly for filter purging and every 3 months for sheath filter replacement.	UN-12	SS-01	A
SYS-1005	Lasers, PMTs, detector arrays, PCBAs and the fluidics supply module shall be field-replaceable by a service engineer.	UN-12	SYS	D
SYS-1006	A trained service engineer shall be able to replace any field-replaceable unit and return the system to passing QC within 4 hours.	UN-12	SYS	D
SYS-1007	The system shall provide diagnostic data (pressures, laser status, voltages, error logs) to service engineers without access to patient data.	UN-12, UN-10	SS-06, SS-10	D
SYS-1008	Lasers and detector arrays shall be upgradable in the field according to CFG-CSI-001.	UN-16	SS-04, SS-05	D

11. Labeling and Documentation Requirements

ID	Requirement	Source	Allocation	Verification
SYS-1101	Product labels shall meet ISO 15223-1 and ISO 18113-3 and shall include the UDI in human-readable and machine-readable form.	UN-13	SYS	I
SYS-1102	The IFU (Operator's Guide) shall include intended use, limitations, warnings, installation requirements, operation, QC, maintenance and troubleshooting.	UN-13, ISO 18113-3	SYS	I
SYS-1103	Laser, high-voltage, biohazard and pinch-point labels shall be placed where the hazard exists, as defined in LBL-CSI-001.	HA-01, HA-02, HA-06	SYS	I
SYS-1104	The system shall be supplied with a site preparation guide and a service manual.	UN-07, UN-12	SYS	I
SYS-1105	Labeling and IFU shall be available in the languages required by each market.	UN-13	SYS	I

12. Packaging, Transport and Installation Requirements

ID	Requirement	Source	Allocation	Verification
SYS-1201	The shipping configuration shall protect the sorter and the fluidics supply module so that, after ISTA 3E transport testing, the system passes installation qualification without optical realignment.	UN-07, UN-08	SYS	T
SYS-1202	Shipping crates shall carry shock and tilt indicators.	UN-12	SYS	I
SYS-1203	Installation, including installation qualification, shall be completed by a service engineer in one working day.	UN-07	SYS	D
SYS-1204	The system shall pass through a standard 90 cm (36 in) laboratory door with covers or crate removed.	UN-07	SYS	I

13. Regulatory and Standards Compliance Requirements

ID	Requirement	Source	Allocation	Verification
SYS-1301	The system shall be designed and manufactured under a quality management system that meets 21 CFR 820 (QMSR) and ISO 13485:2016.	UN-13	SYS	I
SYS-1302	The system shall meet the requirements for US marketing as a Class II IVD instrument through premarket notification (510(k)).	UN-13	SYS	A
SYS-1303	The system shall meet the general safety and performance requirements of EU IVDR 2017/746 (Annex I).	UN-13	SYS	A
SYS-1304	Risk management shall be carried out according to ISO 14971:2019 and all residual risks shall be acceptable under RMP-CSI-001.	UN-06, UN-13	SYS	A
SYS-1305	Software shall be developed according to IEC 62304 and IEC 82304-1.	UN-13	SS-10	I
SYS-1306	The system shall comply with applicable restrictions on hazardous substances (EU RoHS) and with WEEE marking requirements.	UN-13	SYS	I

14. Requirements Traceability

Traceability is maintained in the requirements management tool and exported to the design transfer trace matrix (TM-CSI-DT-01). Each system requirement traces up to its source (user need, hazard or standard) and down to the subsystem specification that implements it and the report that verifies it. Each user need is validated in design validation.

14.1 User needs to system requirements

User need	System requirements	Validation
UN-01 Accurate identification and enumeration	SYS-401 to 445, 495, 632, 704, 806	DVAL-CSI-001 clinical method comparison
UN-02 Pure, viable sorted cells	SYS-461 to 466, 631, 633, 634	DVAL-CSI-001 sort purity and viability study
UN-03 Multicolor panels	SYS-431, 432, 435, 441, 632	DVAL-CSI-001 panel study
UN-04 Practical throughput	SYS-451 to 453, 492, 494, 604	DVAL-CSI-001 workflow study
UN-05 Tubes, plates, slides, single cells	SYS-464, 481 to 485, 631	DVAL-CSI-001 plate sorting study
UN-06 Safe operation	SYS-501 to 563, 803, 804, 902	Usability summative evaluation
UN-07 Standard laboratory installation	SYS-301 to 331, 438, 1104, 1201, 1203, 1204	Site installation at validation sites
UN-08 Consistent daily performance	SYS-436, 437, 444, 465, 468, 613, 621 to 624, 1201	DVAL-CSI-001 multi-day precision
UN-09 Guided, automated procedures	SYS-437, 465, 493, 601 to 614, 805, 901	Usability summative evaluation
UN-10 Secure, traceable records	SYS-635, 701 to 709, 802, 904, 1007	Part 11 assessment; usability evaluation
UN-11 Temperature control	SYS-4101, 4102, 803	DVAL-CSI-001 temperature study
UN-12 Predictable maintenance and repair	SYS-1001 to 1007, 1104, 1202	Service validation at beta sites
UN-13 US and EU compliance	SYS-1101, 1102, 1105, 1301 to 1306	Regulatory submission review
UN-14 No carryover	SYS-471, 472, 603	DVAL-CSI-001 carryover study
UN-15 Different cell sizes and fragility	SYS-423, 463, 467, 491, 611	DVAL-CSI-001 nozzle study
UN-16 Configurable and upgradable	SYS-431, 432, 442, 623, 1008	Configuration demonstration

14.2 Verification documents by section

SRS section	Verification document
3. Physical, electrical and environmental	PVR-CSI-001 §3; SVR-CSI-001 (environmental and electrical)
4. Performance	PVR-CSI-001 §4 to §9; test methods per TMV-CSI-001
5. Safety	SVR-CSI-001 (IEC 61010, 61326, 60825); PVR-CSI-001 §10 (clog and containment); RMR-CSI-001
6. Functional	PVR-CSI-001 §11; software verification per SRS-SW-CSI-010
7. Software, data integrity, cybersecurity	Software verification report; cybersecurity test report; Part 11 assessment
8. Interfaces	ICD-CSI-010 and ICD-CSI-020 verification; PVR-CSI-001 §12
9. Usability	Usability engineering file (formative and summative reports)
10. Reliability and serviceability	Reliability prediction and life test report; service validation report
11. Labeling	Labeling review against LBL-CSI-001
12. Packaging and installation	ISTA 3E transport test report; installation qualification records
13. Regulatory	Regulatory compliance checklist; design history file review

14.3 Requirement allocation summary

Every requirement allocated to a subsystem is decomposed in that subsystem's design specification (Section 2.4). Requirements allocated to SYS are verified at system level and are not decomposed further. No requirement may remain without a source, an allocation and a verification method at Design Input Review 3.

15. Reference Documents

15.1 External standards and regulations

Reference	Title
21 CFR Part 820 (QMSR)	Quality Management System Regulation
21 CFR Part 11	Electronic records; electronic signatures
ISO 13485:2016	Medical devices: quality management systems
ISO 14971:2019	Medical devices: application of risk management to medical devices
IEC 61010-1 / IEC 61010-2-101	Safety requirements for electrical equipment for laboratory use; particular requirements for IVD equipment
IEC 61326-1 / IEC 61326-2-6	EMC requirements for laboratory equipment; particular requirements for IVD equipment
IEC 60825-1	Safety of laser products
IEC 62304 / IEC 82304-1	Medical device software life cycle; health software product safety
IEC 62366-1	Application of usability engineering to medical devices
ISO 15223-1	Symbols to be used with information supplied by the manufacturer
ISO 18113-3	IVD medical devices: information supplied by the manufacturer for IVD instruments for professional use
ISTA 3E	Transport testing of unitized loads
EU IVDR 2017/746	In vitro diagnostic medical devices regulation

15.2 Program documents

Document no.	Rev	Title
PRD-CSI-001	C	Product Requirements Document
SRS-SW-CSI-010	E	Software Requirements Specification: Research, Firmware, Admin and Data Management
SAD-CSI-001	C	System Architecture Description
ICD-CSI-010	B	Interface Control Document: Sorter to Fluidics Supply Module
ICD-CSI-020	A	Interface Control Document: Sorter to Workstation
CFG-CSI-001	C	Configuration Definition: Laser and Detector Builds
SSD-CSI-100	C	Fluidics Subsystem Design Specification
SSD-CSI-300	B	Flow Cell and Nozzle Subsystem Design Specification
SSD-CSI-400	C	Excitation Optics Subsystem Design Specification
SSD-CSI-500	C	Collection and Detection Subsystem Design Specification
SSD-CSI-600	B	Electronics and Signal Processing Subsystem Design Specification
SSD-CSI-700	C	Sort Subsystem Design Specification
SSD-CSI-800	B	Sort Collection and Plate Deposition Subsystem Design Specification
SSD-CSI-900	B	Aerosol Management Subsystem Design Specification
SSD-CSI-950	A	Collection Temperature Control Option Design Specification
HA-CSI-001	C	Hazard Analysis
RMP-CSI-001 / RMR-CSI-001	B / A	Risk Management Plan / Risk Management Report
PVR-CSI-001	A	System Performance Verification Report
SVR-CSI-001	A	Safety, EMC and Laser Compliance Test Report
TMV-CSI-001	A	Test Method Validation Protocols and Reports
DVAL-CSI-001	A	Design Validation Plan and Report
TM-CSI-DT-01	B	Design Transfer Trace Matrix
LBL-CSI-001	B	Labeling Specification
ACC-CSI-001	B	Accessory Kit and Consumables Specification
OG-CSI-001	B	Operator's Guide (Instructions for Use)
DTP-CSI-001	B	Design Transfer Plan

Appendix A. Terms and Abbreviations

Term	Meaning
CEN	Chicken erythrocyte nuclei (resolution and linearity reference particle)
CV	Coefficient of variation
Design input	A physical or performance requirement of the device used as the basis for design (21 CFR 820, ISO 13485 7.3.3)
Drop delay	Time, in drops, from interrogation to breakoff; determines which drop is charged
FCS	Flow Cytometry Standard data file format
FSC / SSC	Forward scatter / side scatter
HA	Hazard identified in the hazard analysis HA-CSI-001
IFU	Instructions for use (Operator's Guide)
PBMC	Peripheral blood mononuclear cells
PMT	Photomultiplier tube
Poisson expected yield	Theoretical maximum yield given the event rate and sort mode
SS-nn	Subsystem identifier (Section 2.4)
UN-nn	User need identifier (Section 2.6)
ULPA	Ultra-low penetration air filter
UDI	Unique device identifier